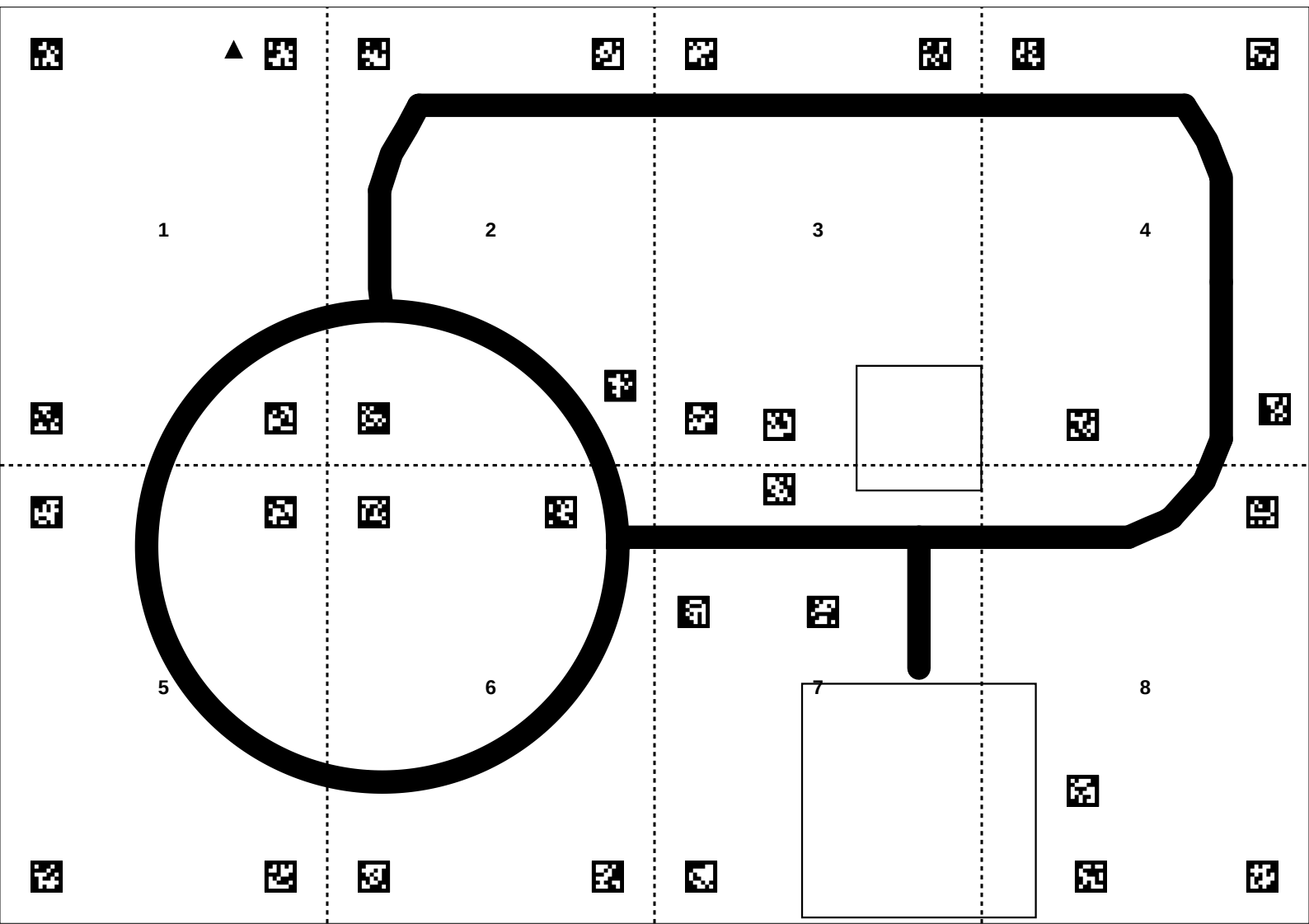


Task-1 Map — Overview

Full-map overview — tile layout reference



Assemble order

Top row (north): tiles 1 – 4, left to right.

Bottom row (south): tiles 5 – 8, left to right.

Align the corner crosses on adjacent tiles and tape on the back.

Task-1 Map — Print & Assemble Guide

1. Print the 8 map tiles (PDF pages 4–11)

Print PDF pages 4–11 on A4 at 100% / Actual Size. Do NOT use “fit to page”.

Verify the scale bar (X 80 mm / Y 40 mm) with a ruler before assembling.

If the corner crosses are clipped, enable borderless printing or reduce printer margins.

2. Assemble (4 columns x 2 rows) — see the overview on PDF page 1

Top row (north) = tiles 1–4 left to right; bottom row (south) = tiles 5–8 left to right.

Align the corner crosses on adjacent tiles; tape on the back.

3. Orientation

The N arrow points map-north (up). Tags are printed with ID upright facing north.

4. AprilTag landmarks (the printed squares)

32 AprilTags (family 36h11, 20 mm) are printed onto the map as absolute-position landmarks.

Each is labelled “ID n (N ↑)” (yaw = 0 = facing map-north).

Source: AprilTag is an open-source project by the University of Michigan APRIL lab.

Official repository: <https://github.com/AprilRobotics/apriltag>

Web search: “AprilTag”, “AprilTag 36h11”, “AprilTag pose estimation”, “OpenCV aruco AprilTag”.

5. Scale

Map 840 x 588 mm = original 1000 x 700 mm Task-1 map scaled 0.84.

Route black line 15 mm; AprilTags 20 mm (physical sizes unchanged).

AprilTag Coordinates

X and Y are in millimetres from the map's SOUTH-WEST corner (X east, Y north); NE corner = (840, 588). cm = mm ÷ 10.

ID	X (mm)	Y (mm)	ID	X (mm)	Y (mm)
0	30.0	558.0	16	30.0	264.0
1	180.0	558.0	17	180.0	264.0
2	30.0	324.0	18	30.0	30.0
3	180.0	324.0	19	180.0	30.0
4	240.0	558.0	20	240.0	264.0
5	390.0	558.0	21	360.0	264.0
6	240.0	324.0	22	240.0	30.0
7	398.0	345.0	23	390.0	30.0
8	450.0	558.0	24	445.0	200.0
9	600.0	558.0	25	500.0	278.5
10	450.0	324.0	26	450.0	30.0
11	500.0	320.0	27	528.0	200.0
12	660.0	558.0	28	695.0	85.0
13	810.0	558.0	29	810.0	264.0
14	695.0	320.0	30	700.0	30.0
15	818.0	330.0	31	810.0	30.0

Same values as scratch/landmarks/task1-tag-map.json (metres = mm ÷ 1000).

tile 1/8

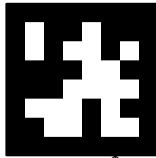


ID 0 (N ↑)

N

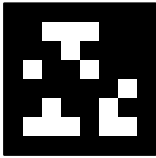


tile 1/8



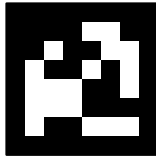
ID 1 (N ↑)

ID 2 (N ↑)



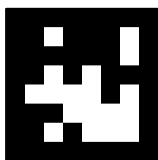
tile 1/8

ID 3 (N ↑)



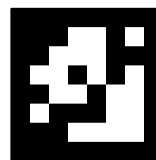
tile 1/8

tile 2/8



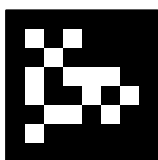
ID 4 (N ↑)

tile 2/8

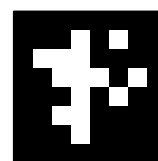


ID 5 (N ↑)

ID 6 (N ↑)



tile 2/8



ID 7 (N ↑)

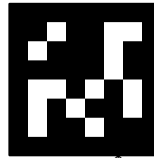
tile 2/8

tile 3/8



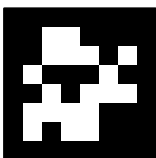
ID 8 (N ↑)

tile 3/8



ID 9 (N ↑)

ID 10 (N ↑)



tile 3/8

ID 11 (N ↑)



scale bar (X 80 / Y 40 mm)

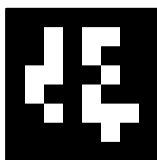
40 mm

0

tile 3/8

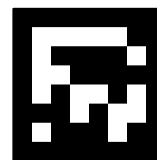
80 mm

tile 4/8



ID 12 (N ↑)

tile 4/8



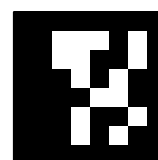
ID 13 (N ↑)

ID 14 (N ↑)



tile 4/8

tile 4/8



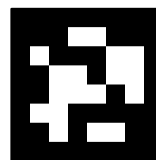
ID 15 (N ↑)

tile 5/8



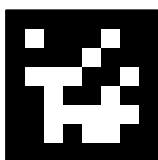
ID 16 (N ↑)

tile 5/8



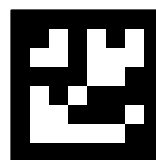
ID 17 (N ↑)

ID 18 (N ↑)



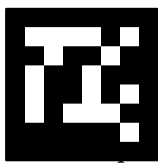
tile 5/8

ID 19 (N ↑)

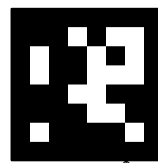


tile 5/8

tile 6/8

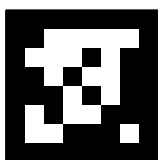


ID 20 (N ↑)



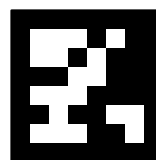
ID 21 (N ↑)

ID 22 (N ↑)

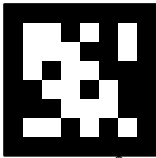


tile 6/8

ID 23 (N ↑)



tile 6/8

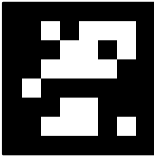


ID 25 (N ↑)

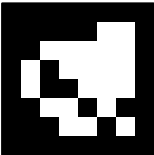
ID 24 (N ↑)



ID 27 (N ↑)

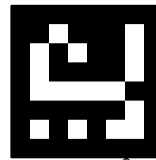


ID 26 (N ↑)



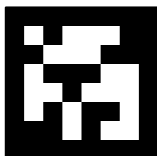
DEPARTURE AREA

START

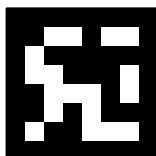


ID 29 (N ↑)

ID 28 (N ↑)



ID 30 (N ↑)



ID 31 (N ↑)

